L4D2 Map Editor: the complete guide

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Tool and guide by **Rune**. Free to use, change and share; please credit Rune if you pass it on.

`L4D2_Map_Editor.py` changes Left 4 Dead 2 maps without recompiling them. It removes invisible walls and edits entities (doors, ladders, blockers, props, triggers, logic), then saves the change as a small `.lmp` patch file you ship in an addon. It works on official maps and custom maps, and keeps the fixes Valve already made.

Tested in game on 2026-11-04: a patch saved this way, shipped in an addon `.vpk`, was loaded by the game over Valve's own patch for c1m2, in co-op and in versus.

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1. Quick start

1. Install Python 3 from python.org (tick **Add Python to PATH**).
2. Open a command prompt in the folder with `L4D2_Map_Editor.py` and run `python L4D2_Map_Editor.py`.
3. Click **Open map...** and pick a `.bsp`, or an addon `.vpk` for a custom map.

4. **Walls** mode: click the red boxes to remove invisible walls. **Entities** mode: select things and delete, move, copy or edit them.
5. Click **Save .lmp**, pick your addon's `maps` folder, confirm. Pack the addon and load the map.

2. Installing and running

- **Needs:** Windows, Python 3.8 or newer. Nothing else to install: it only uses what comes with Python.
- **Run it:** in File Explorer, open the folder with the script, click the address bar, type `cmd`, press Enter, then type `python L4D2_Map_Editor.py`.
- **Double-clicking** may open the script in a code editor (VS Code, Notepad++) instead of running it. That only means `.py` files are set to open there; use the command above.
- **"python is not recognized":** try `py L4D2_Map_Editor.py`. If that fails too, reinstall Python and tick **Add Python to PATH**.
- **Open a map straight away:** `python L4D2_Map_Editor.py "C:\path\to\map.bsp"`.
- The window needs about 1280 x 640 pixels. It opens on a how-to page; **Help** shows it again later.

3. Opening a map

Click **Open map...** and pick one of these:

What	Where to find it
Official maps	`Left 4 Dead 2/left4dead2/maps/`, `left4dead2_dlc1/maps/`, `left4dead2_dlc2/maps/`, `left4dead2_dlc3/maps/` (all loose `.bsp` files)
The Last Stand (c14)	`Left 4 Dead 2/update/maps/`
Workshop maps	`Left 4 Dead 2/left4dead2/addons/workshop/<number>.vpk`
Other custom maps	the addon's `.vpk` in `left4dead2/addons/`, or the map's `.bsp`

- A `.vpk` with several maps shows a list: pick one.

- A split pack (`pak01_000.vpk` , `pak01_001.vpk` …) is opened through its `pak01_dir.vpk` .
- When the map is inside your Left 4 Dead 2 folder, the tool also reads Valve's patches in `update/maps/` for it. For a map outside the game folder it can't, and says so in the grey line at the bottom.
- Compressed maps (LZMA lumps) are supported.

4. Moving around the map

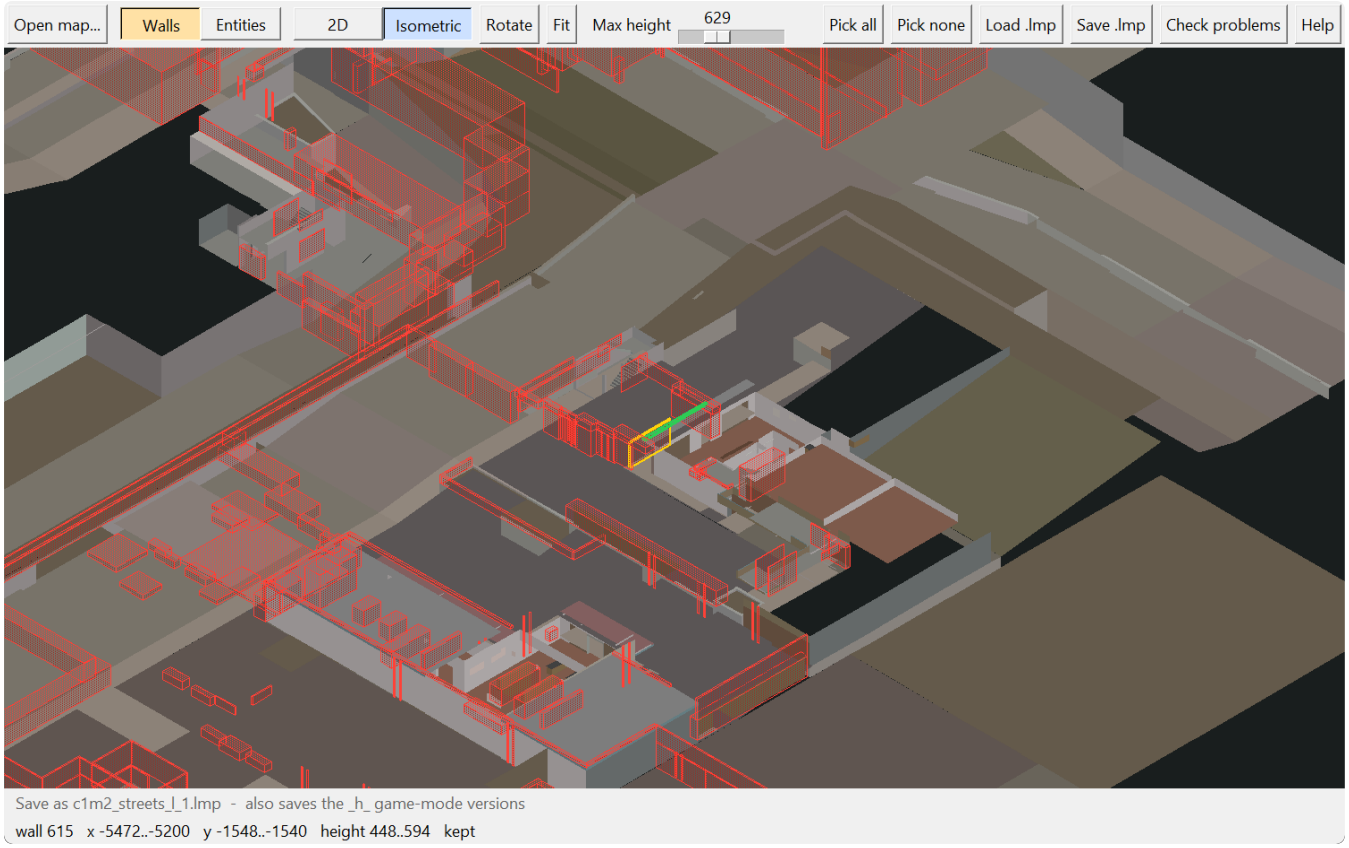
- **2D** shows the map flat from above. **Isometric** shows it in 3D, colored like the textures in game, with simple lighting. Switch any time with the two buttons at the top.
- **Rotate** turns the map 90 degrees. **Fit** shows the whole map again.
- **Max height** cuts away everything above that height. Drag it down to remove roofs and sky walls and see the street level. Your zoom and position stay where they are.
- The grey line at the bottom says where the map's patches come from and what file name Save will use. The black line shows what's under the mouse.

Mouse / key	What it does
Mouse wheel	Zoom in and out, toward the mouse
Right-drag	Move the view
Left click	Walls: remove/keep a wall. Entities: select
Ctrl + click	Entities: add to / remove from the selection
Shift + drag	Pick every wall, or select every entity, inside a box
Drag a selected entity (2D view)	Move it
Arrow keys / Shift + arrows	Move the selection 8 / 64 units
Page Up / Page Down (+ Shift)	Move the selection up / down 8 (64) units
Delete	Delete the selected entities
Ctrl + D	Copy the selected entities
Ctrl + Z / Ctrl + Y	Undo / redo entity changes
Esc	Stop placing a new entity

5. Walls mode

Invisible walls ("player clip") are drawn as red boxes. They block players but are not visible in game.

- **Click** a wall to remove it: it turns green. Click again to keep it.
- **Shift + drag** picks every wall inside the box. **Pick all** picks every wall currently shown (respects the height slider). **Pick none** clears.
- **Hover** a wall: the bottom line shows its number, its X/Y extent and its height range. Walls going up to the sky (for example 592..1472) keep players off roofs; low ones block streets and gaps.
- To match a spot in game: stand there and type `getpos` in the console. Compare the X/Y/Z with the hover info.
- Walls only affect players. Clip that only blocks common infected ("monster clip") stays.



6. Entities mode

Entities are everything placed in the map that isn't solid world: doors, ladders, props, weapon spawns, triggers, blockers, logic (relays, timers), sounds and lights.

Finding things

- The list on the right shows every entity: class and name. **Search** matches class, name, model and hammer ID (several words = all must match). The dropdown filters by category.
- On the map, brush entities (doors, ladders, func_brush…) are boxes and point entities are dots, colored by category:

Category	Color	Examples
Blockers & clips	orange	env_physics_blocker, env_player_blocker, func_playerinfected_clip, func_nav_blocker, func_brush
Ladders	light blue	func_simpleladder
Doors & breakables	blue	prop_door_rotating, func_door, func_breakable, func_button
Props	white	prop_dynamic, prop_physics
Weapons & items	yellow	weapon_*spawn, weapon_spawn, upgrade*
Triggers	purple	trigger_once, trigger_multiple, trigger_hurt, trigger_push
Spawns & nav	green	info_survivor_position, info_changelevel, info_landmark
Logic & other	grey	logic_relay, logic_auto, ambient_generic, env_*

Changing things

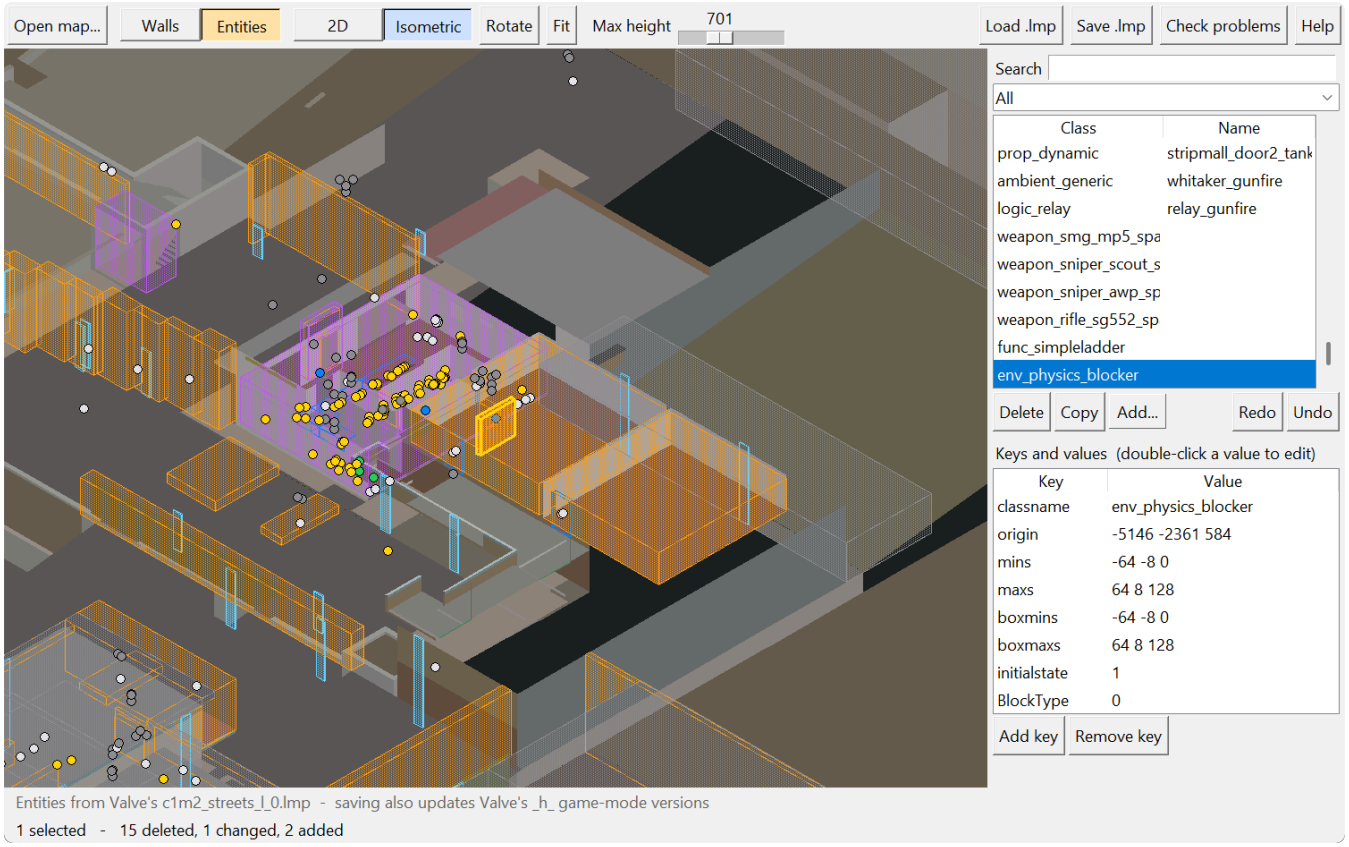
- **Select:** click a dot/box or a row. Ctrl + click adds more; Shift + drag selects a box.
- **Edit a value:** the keys and values of the selected entity show below the list. Double-click a value, type, press Enter. **Add key** / **Remove key** add or remove a line (every entity keeps its `classname`).
- **Outputs** (keys starting with `On`, like `OnTrigger`) show their parts separated by commas: `target,input,parameter,delay,times`. Keep that form when editing.
- **Delete:** Delete key or the **Delete** button. If other entities still send outputs to it, the bottom line warns you.
- **Copy:** Ctrl + D copies the selection 32 units to the side, with new hammer IDs.
- **Move:** drag in the 2D view, or arrow keys / Page Up / Page Down in either view.
- **Add…** places something new where you click next (on a floor or wall you can see):

  - *Invisible wall (blocks everyone)*: an `env_physics_blocker`, 128 wide, 16 deep, 128 tall, sitting on the floor. Change `mins` / `maxs` for another size, `BlockType` for who it blocks.
  - *Concrete barrier prop*: a visible barrier (`prop_dynamic`, solid). Change `model` for another prop.
  - *Copy of the selected entity*: select a ladder, prop or anything else first; the copy lands on the spot you click.
- **Undo / Redo** work for every entity change since you opened the map.

Good to know about brush entities (ladders, doors, func_brush): their shape is part of the map. A copy reuses the same shape, and its `origin` is an offset from where the original stands, not a world position. That's normal.

Recipes

Job	How
Remove a barricade	Search its name (find it by clicking it on the map), select all its parts, Delete. Check the warning for leftover outputs.
Block a path	Add... > Invisible wall, click the floor. Add a visible prop too, so players see why.
Add a ladder over a fence	Select an existing ladder that faces the same way, Add... > Copy of the selected entity, click the spot. Then give bots a nav ladder (see FAQ).
Lock a door shut	Select the `prop_door_rotating` and add 2048 ("starts locked") to its `spawnflags`. For a door nobody should ever pass, delete it and place a barrier prop instead.
Remove a weapon/item spawn	Filter Weapons & items, select, Delete.
Change what spawns	Edit the spawn's `classname` or its weapon keys.
Stop an event from firing	Find the trigger or relay that starts it, delete it or remove its output line.



7. Saving, loading and file numbers

Save .lmp saves the mode you're in (walls or entities). It asks for a folder: pick your addon's `maps` folder (it can be empty or not exist yet). Then it lists the files it will write and asks you to confirm.

File names are `<map>_l_<number>.lmp`. The game loads `_l_0`, `_l_1`, `_l_2` ... in order, stopping at the first missing number, so numbers matter. The tool picks them for you:

- It skips numbers Valve already uses in `update/maps/` (most official maps already have `_l_0`), and numbers already shipped inside the same addon `.vpk`.
- Walls and entities are different patches, so they get different numbers. Save both into the same folder and they won't collide.
- Saving the same kind again into the same folder replaces your earlier file instead of adding a new number.
- If a patch already changes the same thing (Valve patches the walls of c11m3_garage, and the entities of most maps), the tool starts from that patch, so Valve's fix stays in.

Examples:

Map	Walls saved as	Entities saved as
c1m2_streets (Valve has `_l_0` entities, `_h_0`)	`_l_1`, `_h_1`	`_l_2`, `_h_2` if walls were saved first
c11m3_garage (Valve has `_l_0` entities, `_l_1` walls, `_h_0`)	`_l_2`, `_h_1`	next free numbers
a custom map with no patches	`_l_0`	`_l_1`

`_h_` and `_s_` files. Valve ships extra versions of some maps' patches next to `_l_` (`_s_` on the survival maps, `_h_` on 44 maps). In the in-game test, co-op and versus both used the `_l_` files. The tool still writes your change into each version the map has, so it holds wherever the game uses them. If a change can't apply to one version (that version lacks the entity), the save dialog says so.

Load .lmp opens a patch you saved before: walls come back as picked walls, entities come back with all your changes. Edit more, then save again to the same folder to update it.

8. Putting it in an addon

An addon is a folder packed into a `.vpk`:

```
myaddon/
    addoninfo.txt
    maps/
        c1m2_streets_l_1.lmp
        c1m2_streets_h_1.lmp
```

A minimal `addoninfo.txt`:

```
"AddonInfo"
{
    addontitle      "My map fixes"
    addonauthor     "Rune"
    addonversion    1
    addonContent_Map 0
}
```

Pack it: drag the `myaddon` folder onto `Left 4 Dead 2/bin/vpk.exe` (or run `vpk.exe myaddon`). Put `myaddon.vpk` in `left4dead2/addons/`, enable it in the game's Add-ons menu, and load the map. Upload it to the Workshop the usual way.

Everyone playing together should have the same addon. Players without it can rubber-band at walls the others don't have.

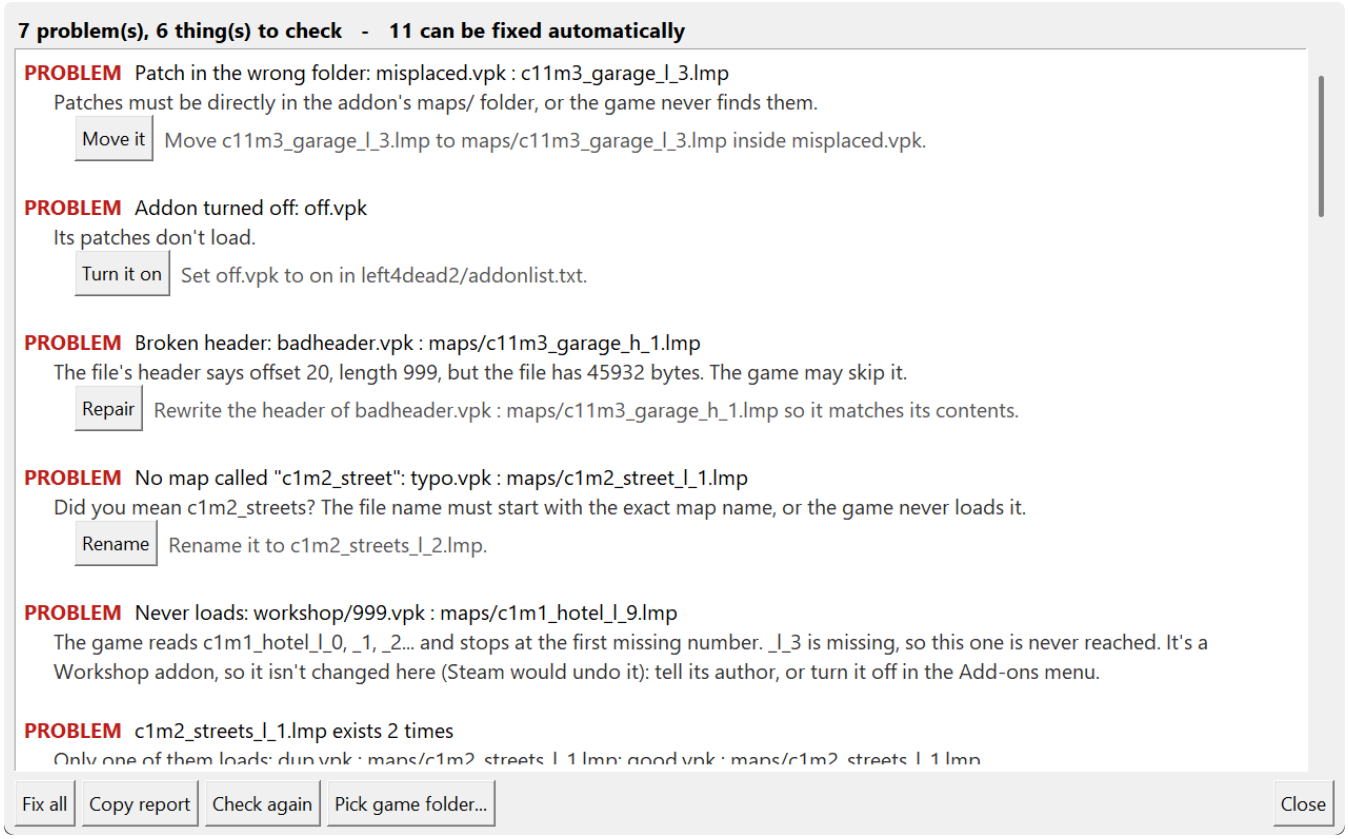
9. Checking it in game

Console command	What it shows
`map c1m2_streets`	Loads a map (patches load with it)
`sv_cheats 1`	Needed for the commands below
`r_drawclipbrushes 2`	Draws invisible walls: removed ones are gone
`getpos`	Your position, to match the tool's coordinates
`ent_bbox env_physics_blocker`	Outlines every blocker of that class (works with any class or name)
`ent_text`	Info on the entity you're looking at (class, name, position)
`noclip`	Fly around to check spots quickly
`nav_edit 1`	Shows the bot navigation mesh

After changing a patch, load the map again: patches only load with the map.

10. Check for problems (and automatic fixes)

Click **Check problems** (top right). The tool finds your Left 4 Dead 2 folder on its own (from Steam's settings), reads every addon and patch, and lists what's wrong in plain words. **Pick game folder...** lets you point it elsewhere. Nothing is changed by checking.



What it looks at

Check	What it means	Automatic fix
Addon turned off	Its patches don't load	Turn it on (addonlist.txt)
Unpacked addon folder	Loose folders in `addons/` don't load	Pack it into a .vpk and turn it on
Patch in the wrong folder	Must be directly in `maps/`	Move it inside the .vpk
Badly named patch	Must be `<map>_l_<number>.lmp`	-
No map with that name	Typo in the map name; suggests the right one	Rename it
Never loads	A number below it is missing: the game stops at the first gap	Renumber it
Same file name twice	Only one of them loads	Renumber one
Replaced by another patch	Two patches change the same thing; only the higher number counts	Merge them into one file and renumber
Drops Valve's own fix	Built on the map without Valve's patch (old tool or other editor)	Rebuild it on Valve's patch, keeping your changes

Check	What it means	Automatic fix
Broken header	The file's header doesn't match its contents	Repair the header
Walls patch doesn't fit	Made for another map or an older version	- (save again from the tool)
Broken entity text / worldspawn not first / entities without class	The map may not load	Fix order / remove bad entities
Model doesn't exist	A prop shows as an error or nothing	-
Blocker with mins bigger than maxs	The box doesn't block	-
New ladders but no .nav	Bots won't climb them	- (see FAQ)
Outputs pointing at deleted entities	Usually fine, just check	-
Patch changes nothing	Same as the map	-
Addon replaces a whole official map	Patches must match that addon's map	-
No addoninfo.txt	Loads, but no title in the Add-ons menu	Add one

Fixing

- Each fixable problem has its own button (**Turn it on**, **Renumber**, **Merge**, **Rebuild**...). **Fix all** fixes everything fixable, including problems that only show up after an earlier fix, and checks again between each step.
- Before anything changes, it lists exactly what it will do and asks.
- Every file is backed up first, in `left4dead2/map_editor_backups/<date-time>/`, keeping the original folder layout. To undo, copy the files back.
- It refuses to fix while Left 4 Dead 2 is running (the game keeps addon files open and rewrites `addonlist.txt` when it closes).
- It never changes Workshop addons (Steam would undo it) or addons split into several .vpk parts. It tells you what to do instead.
- **Copy report** puts the whole report on the clipboard, ready to paste when asking someone for help.

If the tool itself ever hits an error, it says so in plain words and copies the technical details to the clipboard for the same reason.

Tested on a fake game folder with 12 planted problems built from real maps and Valve's real patches: Fix all fixed every fixable one in one go, and checks afterwards confirmed every change survived (both patches' entity edits merged, all 20 walls from 3 files combined, Valve's brush fixes kept).

11. How it works

A Source map (`.bsp`) is split into 64 parts called lumps. A `.lmp` file replaces one lump when the map loads. The game looks for `maps/<map>_l_0.lmp` , `_l_1` , ... in every game folder and addon, and stops at the first missing number. When two files replace the same lump, the higher number wins: that's how this tool's patch overrides Valve's (tested in game).

File layout. 20-byte header of 5 little-endian ints: offset (always 20), lump ID, lump version, length, map revision. The lump data follows. The map revision isn't strictly checked (Valve's own c1m2 patch says 1234 while the map says 21); the tool writes the map's value.

Walls (lump 18, brushes). 12 bytes per brush: first side, number of sides, contents flags. A brush with `0x10000` (PLAYERCLIP) and without `0x1` (SOLID) is an invisible wall. The tool clears `0x10000` on the walls you pick and leaves everything else untouched. Monster clip (`0x20000`) stays. The brush count never changes, so the rest of the map lines up.

Entities (lump 0). Plain text, one block per entity:

```
{
"classname" "func_simpleladder"
"model" "*195"
"origin" "475 -237 176"
"team" "2"
}
```

Keys can repeat (outputs). Output parts are separated by an invisible character (ESC, 0x1B), shown as commas in the tool. The tool writes untouched entities back byte for byte (checked on all 184 entity lumps Valve ships) and only rewrites the ones you changed, in the map's own line-ending style.

Matching entities across versions. Changes are tracked per entity by `hammerid` (or class + name + origin + model when there's none). That's how one edit is applied to Valve's `_l_` , `_h_` and `_s_` versions, which list slightly different entities.

Isometric render. Built from the map's own faces, colored with each texture's average color stored in the map, lit by a fixed sun, drawn back to front. Static props (`prop_static`) aren't part of that data, so they're not drawn.

12. Limits

- It can't change world geometry: no new rooms, walls, floors or holes. Add an invisible wall entity or a prop instead.
- Solid world walls (`SOLID`) can't be removed, only player clip.
- Static props (`prop_static`) can't be edited: they aren't entities.
- Things Valve's scripts add while the map runs (`scripts/vscripts/anv_*.nuc`, from The Last Stand update) aren't in the map's data, so the tool can't see or remove them. In the versus test, those scripts added fences, plywood and "wrong way" walls on c1m2.
- Bots and infected follow the nav mesh, which a `.lmp` can't change. Paths you open or close may need nav edits (see FAQ).
- Only one addon's `<map>_l_<number>.lmp` with the same name loads.

13. Troubleshooting and FAQ

Start here: click **Check problems**. It catches most of the cases below on its own and can fix many of them. If you ask someone for help, send them the report (**Copy report**).

Why use this instead of Hammer? Hammer builds maps; this patches existing ones. No decompiling (BSPSource never gives a perfect copy), no recompiling, a few hundred KB instead of a 20-90 MB map, Valve's fixes stay in, fewer conflicts and no "map differs from server". Hammer is still the tool for changing geometry, lighting or textures.

The wall is still there in game.

1. Is the addon enabled, and is the file in its `maps` folder with the exact map name? Load the map again after changes.
2. Another addon with the same file name? Only one loads. Rename or merge (see below).
3. Was it really an invisible wall? Check with `r_drawclipbrushes 2`. If nothing is drawn there, it's an entity: try `ent_bbox env_physics_blocker`, `ent_bbox env_player_blocker`, `ent_bbox func_brush`, then remove it in Entities mode.
4. Still nothing? It may be added at runtime by Valve's scripts (see Limits), or be a solid world wall.

My entity change doesn't show. Load the map again. Make sure you saved in Entities mode (Walls mode saves only walls). Check the file number isn't taken by another addon.

Survivors get through, then die or fall. You opened a way out of the playable area: there may be a `trigger_hurt` or no floor. Look for purple triggers there.

Bots won't follow / get stuck. The nav mesh doesn't know your change. In game: `sv_cheats 1`, `nav_edit 1`, aim at the new path or ladder, `nav_build_ladder` for ladders (or mark/connect areas), then `nav_save`. Ship the `maps/<map>.nav` it writes in your addon.

The ladder copy doesn't work. Pick a source ladder that faces the same way; keep its `normal.x/y/z`; `team` 2 = survivors only, 0 = everyone. Its `origin` is an offset, that's normal.

A placed thing floats or sinks. Click a floor you can see. Fix with Page Up / Page Down, or edit `origin`.

A prop is invisible or an error model. The `model` path must exist in the game. Copy the path from a prop already in the map.

The save dialog warns about outputs pointing at deleted entities. Something still tries to trigger what you deleted. The game ignores those outputs; delete or edit them if the event should change too.

Two addons patch the same map. Load one addon's patch with Load .lmp, redo the other's changes, save one combined file.

Valve updated the map. Open it again and redo the patch (Load .lmp helps): patches are built on the map as it was.

Does it work online / on servers? The patch loads like any addon file. Servers and players should run the same addon.

"Can't read this map". Not a Source map, a damaged file, or a split `.vpk` opened by a part file instead of `pak01_dir.vpk`.

"This map has no invisible walls to remove". It has no player clip brushes; walls there are entities. Use Entities mode.

Is my original map changed? Never. Removing the addon brings the map back.

14. Command line

For scripts and batch jobs (walls only, writes only the `_l_` file):

```
python L4D2_Map_Editor.py check
python L4D2_Map_Editor.py check "D:\SteamLibrary\steamapps\common\Left 4 Dead
2"
python L4D2_Map_Editor.py <map>.bsp write <map>_l_1.lmp all
python L4D2_Map_Editor.py <map>.bsp write <out>.lmp box X0 Y0 X1 Y1
python L4D2_Map_Editor.py <map>.bsp write <out>.lmp 614 615 617
python L4D2_Map_Editor.py <map>.bsp list X0 Y0 X1 Y1
python L4D2_Map_Editor.py <addon>.vpk <map> write <map>_l_0.lmp all
```

`check` prints the same report as the button (no fixes from the command line).

It warns if the file number is already used by one of Valve's patches.